

DERIVATION OF THE STRUCTURAL WAVELENGTHS λ_1 AND λ_2

From the Symmetric Two-Body Free-Fall Relation
to the Closed Complex Branch

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The purpose of this derivation is to reconstruct the closed-form expressions for the structural wavelengths λ_1 and λ_2 from the relations obtained at the end of the two-body free-fall derivation in *The Theory of Symmetric Inertia Transfer*.

The key idea is simple even though the final formulas look complicated. The earlier derivation gives two pieces of information: a gravitational wavelength-product relation and a symmetric mass-wavelength relation. The negative sign in the mass-wavelength relation produces the complex factor i . Once one wavelength is written in terms of the other, that relation can be substituted back into the gravitational branch equation and solved algebraically. The long expressions for λ_1 and λ_2 are therefore just a factorized form of a much simpler cubic solution.

1. Starting Relations from the Two-Body Derivation

For the reciprocal branch associated with Body 1, the closed-form solution carries forward the wavelength-product relation

$$2M_1GT_1^2 = \lambda_1^2(\lambda_2 - \lambda_1).$$

Using the Planck-Einstein relation

$$E = hf = M_1c^2,$$

the period of the first branch is

$$T_1 = \frac{h}{M_1c^2}, \quad T_1^2 = \frac{h^2}{M_1^2c^4}.$$

Substituting this into the wavelength-product relation gives

$$\frac{2Gh^2}{M_1c^4} = \lambda_1^2(\lambda_2 - \lambda_1).$$

The symmetric two-body derivation also produced the mass-wavelength constraint

$$M_1\lambda_1^2 = -M_2\lambda_2^2.$$

This second equation is what introduces the complex branch.

2. Isolating λ_2 in Terms of λ_1

Take the square root of the mass-wavelength relation:

$$\sqrt{M_1 \lambda_1^2} = \sqrt{-M_2 \lambda_2^2}.$$

Because $\sqrt{-1} = i$,

$$\sqrt{M_1} \lambda_1 = \pm i \sqrt{M_2} \lambda_2.$$

The two signs correspond to the two conjugate complex orientations. The closed-form branch used in the manuscript is obtained by choosing

$$\sqrt{M_1} \lambda_1 = -i \sqrt{M_2} \lambda_2.$$

Solving for λ_2 gives

$$\lambda_2 = i \sqrt{\frac{M_1}{M_2}} \lambda_1.$$

This is the crucial substitution. It says that the second structural wavelength is not independent of the first: it is fixed by the mass ratio and a quarter-cycle complex rotation.

3. Substituting into the Gravitational Branch Equation

Return to

$$\frac{2Gh^2}{M_1 c^4} = \lambda_1^2 (\lambda_2 - \lambda_1).$$

Insert

$$\lambda_2 = i \sqrt{\frac{M_1}{M_2}} \lambda_1.$$

Then

$$\frac{2Gh^2}{M_1 c^4} = \lambda_1^2 \left(i \sqrt{\frac{M_1}{M_2}} \lambda_1 - \lambda_1 \right).$$

Factor out λ_1 from the parenthesis:

$$\frac{2Gh^2}{M_1 c^4} = \lambda_1^3 \left(i \sqrt{\frac{M_1}{M_2}} - 1 \right).$$

Therefore,

$$\lambda_1^3 = \frac{2Gh^2}{M_1 c^4} \left(i\sqrt{\frac{M_1}{M_2}} - 1 \right)^{-1}.$$

Taking the cube root gives the compact closed form

$$\lambda_1 = \left(\frac{2Gh^2}{M_1 c^4} \right)^{1/3} \left(i\sqrt{\frac{M_1}{M_2}} - 1 \right)^{-1/3}.$$

This is already a complete solution for λ_1 . The longer expression used in the manuscript is an algebraic refactorization of this same result.

4. Rewriting λ_1 in the Symmetric Factorized Form

Define

$$D \equiv \frac{2Gh^2}{c^4}, \quad Q \equiv \frac{i}{\sqrt{M_2}} - \frac{1}{\sqrt{M_1}}.$$

Notice that

$$i\sqrt{\frac{M_1}{M_2}} - 1 = \sqrt{M_1} \left(\frac{i}{\sqrt{M_2}} - \frac{1}{\sqrt{M_1}} \right) = \sqrt{M_1} Q.$$

The compact solution can therefore be written as

$$\lambda_1 = \left(\frac{D}{M_1} \right)^{1/3} (\sqrt{M_1} Q)^{-1/3}.$$

Combining the powers of M_1 gives

$$\lambda_1 = D^{1/3} M_1^{-1/2} Q^{-1/3}.$$

Now use

$$\frac{h}{c^2} \sqrt{2G} = D^{1/2},$$

and

$$\left[D^{1/3}Q^{2/3}\right]^{-1/2} = D^{-1/6}Q^{-1/3}.$$

Multiplying these factors gives

$$D^{1/2}D^{-1/6} = D^{1/3},$$

so the compact solution becomes exactly

$$\lambda_1 = \frac{h}{c^2} \sqrt{\frac{2G}{M_1}} \left[\left(\frac{2Gh^2}{c^4} \right)^{1/3} \left(\frac{i}{\sqrt{M_2}} - \frac{1}{\sqrt{M_1}} \right)^{2/3} \right]^{-1/2}.$$

The apparently complicated structure is therefore only a symmetric factorization of the simpler cubic-root solution from the previous section.

5. Deriving λ_2

The second wavelength now follows directly from the complex branch relation

$$\lambda_2 = i \sqrt{\frac{M_1}{M_2}} \lambda_1.$$

Substituting the factorized expression for λ_1 gives

$$\lambda_2 = i \sqrt{\frac{M_1}{M_2}} \frac{h}{c^2} \sqrt{\frac{2G}{M_1}} \left[\left(\frac{2Gh^2}{c^4} \right)^{1/3} \left(\frac{i}{\sqrt{M_2}} - \frac{1}{\sqrt{M_1}} \right)^{2/3} \right]^{-1/2}.$$

The mass factors simplify because

$$\sqrt{\frac{M_1}{M_2}} \sqrt{\frac{1}{M_1}} = \frac{1}{\sqrt{M_2}}.$$

Therefore,

$$\lambda_2 = i \frac{h}{c^2} \sqrt{\frac{2G}{M_2}} \left[\left(\frac{2Gh^2}{c^4} \right)^{1/3} \left(\frac{i}{\sqrt{M_2}} - \frac{1}{\sqrt{M_1}} \right)^{2/3} \right]^{-1/2}.$$

This is the second closed structural wavelength used in the theory.

6. Why the Two Final Expressions Have the Same Internal Bracket

Both wavelengths contain the same common structural factor

$$\mathcal{B}(M_1, M_2) = \left[\left(\frac{2Gh^2}{c^4} \right)^{1/3} \left(\frac{i}{\sqrt{M_2}} - \frac{1}{\sqrt{M_1}} \right)^{2/3} \right]^{-1/2}.$$

The two solutions can therefore be displayed transparently as

$$\lambda_1 = \frac{h}{c^2} \sqrt{\frac{2G}{M_1}} \mathcal{B}(M_1, M_2),$$

$$\lambda_2 = i \frac{h}{c^2} \sqrt{\frac{2G}{M_2}} \mathcal{B}(M_1, M_2).$$

Their ratio is immediately

$$\frac{\lambda_2}{\lambda_1} = i \sqrt{\frac{M_1}{M_2}}.$$

Squaring this relation returns the original mass-wavelength constraint:

$$M_2 \lambda_2^2 = M_2 \left(i^2 \frac{M_1}{M_2} \right) \lambda_1^2 = -M_1 \lambda_1^2,$$

or

$$M_1 \lambda_1^2 = -M_2 \lambda_2^2.$$

The closed forms therefore preserve the same complex symmetry from which they were constructed.

7. Derivation in One Chain

The entire calculation can be summarized as

$$M_1 \lambda_1^2 = -M_2 \lambda_2^2$$

↓

$$\lambda_2 = i\sqrt{\frac{M_1}{M_2}}\lambda_1$$

$$\Downarrow$$

$$\frac{2Gh^2}{M_1c^4} = \lambda_1^2(\lambda_2 - \lambda_1) = \lambda_1^3 \left(i\sqrt{\frac{M_1}{M_2}} - 1 \right)$$

$$\Downarrow$$

$$\lambda_1 = \left(\frac{2Gh^2}{M_1c^4} \right)^{1/3} \left(i\sqrt{\frac{M_1}{M_2}} - 1 \right)^{-1/3}$$

$$\Downarrow$$

$$\lambda_1 = \frac{h}{c^2} \sqrt{\frac{2G}{M_1}} \left[\left(\frac{2Gh^2}{c^4} \right)^{1/3} \left(\frac{i}{\sqrt{M_2}} - \frac{1}{\sqrt{M_1}} \right)^{2/3} \right]^{-1/2}$$

and finally

$$\lambda_2 = i\sqrt{\frac{M_1}{M_2}}\lambda_1$$

which gives

$$\lambda_2 = i\frac{h}{c^2} \sqrt{\frac{2G}{M_2}} \left[\left(\frac{2Gh^2}{c^4} \right)^{1/3} \left(\frac{i}{\sqrt{M_2}} - \frac{1}{\sqrt{M_1}} \right)^{2/3} \right]^{-1/2}.$$

The essential move is exactly the one remembered from the original derivation: **solve the symmetric mass-wavelength relation for λ_2 , substitute it into the gravitational equation for λ_1 , solve the resulting cubic, and then recover λ_2 from the symmetry relation.**

Complex-root branch note. The square root and fractional powers are multivalued in the complex plane. The displayed formulas consistently use the branch satisfying $\lambda_2 = i\sqrt{M_1/M_2}\lambda_1$; the conjugate branch follows the same algebra with the opposite complex orientation.